Exact Singularities of Homogeneous Ricci Flow on Products of Round Spheres

Route-A source-local certificate HCS-C281
Revision round 1

1 September 2026

Abstract

For arbitrary $m \geq 1$, dimensions $d_i \geq 1$, and scales $a_i > 0$, we solve the unnormalized Ricci flow of the product metric $\bigoplus_i a_i g_{S^{d_i}}$ exactly. Every tied first-collapse factor is retained, yielding sharp scalar-curvature, full-curvature, volume, diameter, and Type-I blowup laws. We derive the constant-volume conjugacy and prove that partial collapse has finite normalized lifetime, whereas full collapse is exactly the curved Einstein face and becomes a stationary forward-eternal normalized solution. This is a nonlinear metric-evolution theorem, not a heat-trace, spectral-zeta, or determinant construction.

1 Frozen geometric owner

Let

$$M = \prod_{i=1}^{m} S^{d_i}, \qquad d_i \in \mathbb{Z}_{\geq 1}, \qquad g(0) = \bigoplus_{i=1}^{m} a_i g_i, \quad a_i > 0, \tag{1}$$

where g_i is the unit round metric. Put $c_i = d_i - 1$ and $n = \sum_i d_i$. We use the unnormalized equation

$$\partial_t g = -2\,\mathrm{Ric}(g) \tag{2}$$

with its physical clock. Hamilton's original program fixes (2) as a geometric evolution equation [1]; Chow–Knopf provide standard special-geometry, scaling, and singularity conventions [2]. Those sources are cited only for equation lineage and vocabulary. All product, endpoint, and normalization claims below are derived here.

For $d_i \geq 2$ define $T_i = a_i/(2c_i)$; for $d_i = 1$ define $T_i = \infty$. This convention is essential: a circle is Ricci-flat, not a factor with an undefined clock.

Theorem 1 (exact flow and maximal interval)**.** *The solution of (2) is*

$$g(t) = \bigoplus_i a_i(t) g_i, \qquad a_i(t) = a_i - 2c_i t. \tag{3}$$

If some $c_i > 0$, its maximal Riemannian interval is $(-\infty, T)$, where $T = \min_i T_i$. If every $d_i = 1$, (3) is stationary for all $t \in \mathbb{R}$.

Proof. As a $(0,2)$ tensor, the Ricci tensor is unchanged by constant metric rescaling. The unit round factor satisfies $\mathrm{Ric}(g_i) = c_i g_i$, and the Ricci tensor of a Riemannian product is the direct sum of factor tensors. Hence (2) is exactly $a_i' = -2c_i$, proving (3). The coefficients remain positive precisely on the stated interval. The displayed solution solves the full equation; compact Ricci-flow uniqueness identifies it with the ambient solution while it exists. □

2 Closed geometric observables

Mixed sectional curvatures vanish, while the ith factor has sectional curvature $a_i(t)^{-1}$. Consequently

$$R(t) = \sum_i \frac{d_i c_i}{a_i(t)}, \qquad |\mathrm{Ric}|^2(t) = \sum_i \frac{d_i c_i^2}{a_i(t)^2}, \tag{4}$$

$$|\mathrm{Rm}|^2(t) = \sum_i \frac{2 d_i c_i}{a_i(t)^2}, \qquad \frac{V(t)}{V(0)} = \prod_i \left(\frac{a_i(t)}{a_i}\right)^{d_i/2}. \tag{5}$$

Here $V(t) = \mathrm{Vol}(M, g(t))$. Differentiation gives

$$\frac{d}{dt}\log V(t) = -R(t). \tag{6}$$

The product distance also gives the exact diameter

$$\mathrm{diam}(M, g(t)) = \pi\sqrt{\sum_i a_i(t)}. \tag{7}$$

The initial product is Einstein with positive constant λ exactly when every factor is curved and $c_i/a_i = \lambda$ for every i. Equivalently, all finite clocks coincide at $T = (2\lambda)^{-1}$, and then

$$g(t) = (1 - t/T)g(0). \tag{8}$$

If all factors are circles, the product is instead Ricci-flat for arbitrary scales. A mixture of flat and curved factors cannot be Einstein.

3 Tied collapse and the full Type-I model

Assume $T < \infty$ and freeze the complete first-collapse set

$$I = \{i : T_i = T\}, \qquad D = \sum_{i\in I} d_i. \tag{9}$$

No generic perturbation is used to split a tie. For $i \in I$, $a_i(t) = 2c_i(T - t)$; for $j \notin I$, $a_j(T) > 0$.

Theorem 2 (sharp singularity atlas)**.** *As $t \uparrow T$,*

$$(T - t)R(t) \longrightarrow \frac{D}{2}, \tag{10}$$

$$(T - t)^2|\,\mathrm{Ric}\,|^2(t) \longrightarrow \frac{D}{4}, \tag{10a}$$

$$(T - t)^2|\,\mathrm{Rm}\,|^2(t) \longrightarrow \sum_{i\in I}\frac{d_i}{2c_i}, \tag{11}$$

$$V(t) \sim C_I(T - t)^{D/2}, \tag{12}$$

where

$$C_I = \prod_i \mathrm{Vol}(S^{d_i}, g_i)\prod_{i\in I}(2c_i)^{d_i/2}\prod_{j\notin I} a_j(T)^{d_j/2}.$$

The singularity is Type I and

$$\mathrm{diam}(M, g(t)) \longrightarrow \pi\sqrt{\sum_{j\notin I} a_j(T)}. \tag{13}$$

For any $t_k \uparrow T$, let $Q_k = (T - t_k)^{-1}$ and $g_k(s) = Q_k g(T + s/Q_k)$ for $s < 0$. At product basepoints,

$$(M, g_k(s)) \longrightarrow \prod_{i\in I}\big(S^{d_i}, -2c_i s g_i\big) \times (\mathbb{R}^{n-D}, g_{\mathrm{E}}) \tag{14}$$

smoothly on compact pointed sets.

Proof. Substitution of $a_i(t) = 2c_i(T - t)$ into (4)–(5) proves (10), (10a)–(12); every noncollapsing summand stays bounded. Formula (11) gives both the Type-I upper bound and a nonzero Type-I residue, while (7) gives (13). In the rescaled flow, a collapsing coefficient is exactly $-2c_i s$. A survivor has coefficient $Q_k a_j(T) - 2c_j s$, which diverges. A round factor with scale tending to infinity converges pointed smoothly to its Euclidean tangent space. Taking products proves (14). ∎

4 Constant-volume conjugacy and endpoint theorem

Define

$$C(t) = \left(\frac{V(0)}{V(t)}\right)^{2/n}, \qquad \tau(t) = \int_0^t C(s)\,ds, \qquad \widehat{g}(\tau(t)) = C(t)g(t). \tag{15}$$

Theorem 3 (partial versus full collapse)**.** *The metric $\widehat{g}$ has volume $V(0)$ and satisfies*

$$\partial_\tau \widehat{g} = -2\,\mathrm{Ric}(\widehat{g}) + \frac{2}{n}\widehat{R}\,\widehat{g}. \tag{16}$$

It is ancient in normalized time. At the forward endpoint:

1. *If $D = n$, then every factor is curved, (1) is Einstein, (8) holds, $C(t) = (1 - t/T)^{-1}$, and $\tau = -T\log(1 - t/T)$. Thus $\widehat{g} \equiv g(0)$ and the normalized lifetime is infinite.*
2. *If $D < n$, then $\tau_T = \lim_{t\uparrow T}\tau(t) < \infty$. Every normalized collapsing scale tends to zero, every survivor scale tends to infinity, and normalized curvature blows up.*

All-flat products form the separate stationary eternal case $C = 1$, $\tau = t$.

Proof. Equation (15) makes $C^{n/2}V = V(0)$. From (6), $C'/C = 2R/n$. Constant rescaling leaves Ric fixed as a $(0,2)$ tensor and gives $\widehat{R} = R/C$. Dividing $d(Cg)/dt$ by $d\tau/dt = C$ yields (16).

At backward infinity, if D_c is the total dimension of all curved factors, then $C(t)$ is comparable to $|t|^{-D_c/n}$. Its integral diverges for $D_c < n$, and diverges logarithmically for $D_c = n$; hence normalized time tends to $-\infty$. At T, (12) gives $C(t) \asymp (T-t)^{-D/n}$. This is integrable exactly when $D < n$. If $D = n$, all clocks coincide and the Einstein calculation gives the explicit stationary normalization. If $D < n$, multiplication of collapsed affine scales and surviving positive scales by $C(t)$ proves the asserted limits. $\square$

References

[1] Richard S. Hamilton, *Three-manifolds with positive Ricci curvature*, Journal of Differential Geometry 17 (1982), 255–306, doi:10.4310/jdg/1214436922.

[2] Bennett Chow and Dan Knopf, *The Ricci Flow: An Introduction*, Mathematical Surveys and Monographs 110, American Mathematical Society, 2004, doi:10.1090/surv/110.